DS3 v1.0 User Guide

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1. Introduction

DS3 is a desktop application for viewing, correcting and analysing hyperspectral image cubes. Version 1.0 brings loading, preprocessing, radiometric and atmospheric correction, denoising, endmember extraction, unmixing and classification into one window, with a single processing history you can save or undo.

What is new in v1.0

- **Any file format:** ENVI, MATLAB, HDF5/HDF4, NetCDF, Zarr, GeoTIFF, NumPy, FITS and other GDAL formats open from one dialog, with a progress bar in the status bar.
- **Revert and history on every tab:** a bar above the tabs holds Revert to original cube, Show history and Save history.
- **Band Management for any format:** wavelengths can be assigned from a file, a uniform range or a pasted list when the file has none.
- **Data Correction tab:** radiometric calibration, four atmospheric methods (Empirical Line, Dark Pixel Subtraction, Flat Field, IARR), smile reduction and NGmeet denoising, in four equal panels with collapsible descriptions.
- **Spectral Analysis tab:** Band Math, k-means clustering and SNR on one tab.
- **Unmixing tab:** HFC / NWHFC endmember count, N-FINDR, PPI and Fast Iterative PPI extraction, linear unmixing and NMF, in four columns that follow the workflow.

Who this guide is for

Anyone running a demo or analysis session with DS3. Each tab has its own section; section 3 is a ten-step first session; section 14 collects typical workflows.

2. Installation and requirements

DS3 is a single Python file, `HyperVista_v1_0.py`, that runs on Python 3 with the packages below.

Package	Needed for	Required?
numpy, scipy, matplotlib	all numerical work and plots	Yes
PyQt5	the user interface	Yes
spectral	ENVI files and ENVI export	Yes
scikit-image	NGmeet denoising (non-local means, PSNR, SSIM)	For denoising
scikit-learn	k-means, classifiers, NMF	For those tabs
tensorflow	2D CNN classification	Only for the CNN tab
h5py, netCDF4, pyhdf, zarr, astropy	HDF5, NetCDF, HDF4, Zarr, FITS files	Only for those formats
rasterio or GDAL, tifffile	GeoTIFF and other raster formats	Only for those formats

Install the packages you need with pip, for example `pip install numpy scipy matplotlib PyQt5 spectral scikit-image scikit-learn`. Start the application with `python HyperVista_v1_0.py`.

If a package is missing, the matching feature shows a message instead of running; the rest of the application keeps working. Help → Supported file formats lists what your installation can open.

3. Quick start

A first session takes ten steps, from opening a file to saving the result.

1. Choose File → Open hyperspectral image (Ctrl+O) or drag a file onto the window. A progress bar appears in the status bar while the file loads.
2. If the file holds several datasets, or the spectral axis is ambiguous, pick the right one in the dialog that appears.
3. Read the info panel on the right. It shows the cube size and the wavelength range, or says that no wavelengths were found.
4. If there are no wavelengths, open Band Management and use "0. Wavelengths" to assign them.

5. Move the band slider, or switch the Display box to RGB composite, to look at the scene.
6. Click a pixel in the image to see its spectrum in the plot on the right.
7. Correct the data on the Preprocessing or Data Correction tab, for example radiometric calibration, then an atmospheric method.
8. Draw regions on the ROI Analysis tab and add them to the ROI list to build classes.
9. Analyse on the Spectral Analysis, Unmixing, PCA or Classification tabs.
10. Choose Show history to review what was done, then export the cube from the Preprocessing tab or save the history.

At any point, **Revert to original cube** (above the tabs) returns the cube to the state it was loaded in.

4. Interface overview

The window has a viewing area on top, a history bar in the middle and nine tabs at the bottom.

Viewing area

- **Image panel (top left):** shows the current band, or an RGB composite. A slider below it moves through the bands, and the band number is shown beside it.
- **Display box:** choose Single band or RGB composite, and set the R, G and B band numbers.
- **Spectrum plot (top right):** click any pixel in the image to plot its spectrum. Plots use wavelength in nm when the cube has wavelengths, and band number otherwise.
- **Info panel (below the plot):** messages about what the app just did, such as load results, warnings and counts.

History bar (above the tabs, always visible)

- **Revert to original cube:** discards every processing step on every tab and restores the cube as loaded.
- **Show history:** lists every operation applied, with time, parameters and band counts.
- **Save history...:** writes the same list to a JSON or text file.

Menus

Menu	Items
File	Open hyperspectral image (Ctrl+O); Open Zarr folder; Import ASD field spectra; Save ROIs; Load ROIs
View	Theme (Dark or Light); Processing history; Revert to original cube
Help	About DS3; Supported file formats

Tabs, left to right

Tab	What it does
Preprocessing	Normalisation, unit conversion, common wavelength grid, spectral indices, statistics, export
Band Management	Assign wavelengths, remove bands, resample bands, select informative bands
Data Correction	Radiometric calibration, atmospheric correction, smile reduction, denoising
ROI Analysis	Draw regions, manage classes, anomaly map, ROI spectra
Spectral Analysis	Band math, k-means clustering, signal-to-noise ratio
Unmixing	Endmember count, extraction, linear unmixing, NMF
PCA	Principal component analysis and Spectral Angle Mapper
Classification	Supervised classification and cross-validation on labelled ROIs
2D CNN	Deep-learning classification on PCA patches

While a long task runs (denoising, training, endmember extraction), the Stop task button appears in the status bar.

5. Loading data

Use File → Open hyperspectral image (Ctrl+O), or drag a file onto the window. The file dialog opens on the folder you used last.

Supported formats

Format	Typical extensions	Extra package
ENVI (open the header or the data file)	.hdr, .img, .dat, .raw, .bil, .bip, .bsq	none
MATLAB v5 to v7.3	.mat	h5py for v7.3
HDF5 / HDF-EOS5 (PRISMA, DESIS)	.h5, .hdf5, .he5	h5py
HDF4 / HDF-EOS (Hyperion L1R)	.hdf, .h4, .l1r	pyhdf
NetCDF (EMIT, PACE)	.nc, .nc4	netCDF4
GeoTIFF / TIFF (EnMAP)	.tif, .tiff	rasterio, GDAL or tifffile
NumPy	.npy, .npz	none
FITS	.fits	astropy
ERDAS	.lan, .gis	none
Zarr (zipped store or folder)	.zip, .zarr	zarr
IDL save files	.sav	none
Other GDAL rasters	.vrt, .jp2, .ntf, .cub and more	GDAL

For a Zarr folder, use File → Open Zarr folder. ASD point spectra are not image cubes; see the next part.

What happens on load

- A progress bar in the status bar shows the steps. For ENVI files it follows the real read; for other formats it moves through opening, checking and setting up the viewer.
- If a file contains several datasets, a dialog asks which one to open.
- If a 3-D array does not say which axis holds the bands, a dialog asks you to choose among (bands, rows, columns), (rows, bands, columns) or (rows, columns, bands).

- Bands that are entirely NaN are dropped and listed in a message.
- Large cubes are stored as float32 to save memory.
- Loading a new file clears the previous analysis state, ROI references and history.

Wavelengths

The info panel states the wavelength range after loading, or "No wavelength metadata found". Cubes without wavelengths still open and display by band number. To use wavelength-based tools, assign wavelengths in Band Management (section 7).

ASD field spectra

Choose File → Import ASD field spectra to read `.asd` files. The dialog offers splice (detector jump) correction at about 1000 nm and 1800 nm, and, if a cube with wavelengths is loaded, adds the spectra to the endmember list resampled to the cube's bands. Imported spectra can be plotted or saved as CSV from the Unmixing tab.

6. Preprocessing tab

The Preprocessing tab changes the values or wavelengths of the whole cube, one step at a time. It has five pages, chosen with the sub-tabs, and a footer shared by all of them.

Normalisation

Pick a method and click Apply Normalisation.

Method	What it does
Min-max to range	Rescales to a low and high value you set. Scope can be per band, per pixel spectrum or the whole cube; Robust uses the 2nd and 98th percentiles and clips
SNV	Standard normal variate: each pixel spectrum gets mean 0 and standard deviation 1
Z-score	Each band is standardised across the image
Unit vector (L2)	Each pixel spectrum is divided by its length

Units

- **Convert Data Values:** reflectance (scaled ×10000, percent, 0–1) and radiance unit conversions, or a custom value × factor + offset.

- **Convert Wavelengths to nm:** fixes wavelengths stored in micrometres, Å, metres or wavenumbers (cm^{-1}). Wavenumbers are converted and the bands are re-ordered to increasing wavelength.

Common Grid

Resamples every pixel spectrum onto one wavelength grid, either uniform (start, end, step in nm) or read from a file (`.hdr` , `.txt` or `.csv`). Use Cube Range fills the start and end from the cube. Set the FWHM if the grid file does not carry it.

Indices & Features

Computes an image from chosen wavelengths. Types are normalised difference, band ratio, band difference, band depth (continuum removed) and band gap. Presets cover NDVI, NDRE, two NDWI variants, MNDWI, NDBI, NDII, NBR, PRI, simple ratio, iron oxide ratio, chlorophyll, water, Al-OH clay and carbonate depths, and red-edge position. Buttons: Compute Feature Image, Colour Bar & Histogram, Save Image.

Statistics

- **Per-band statistics:** plot mean, standard deviation and range, or save them as CSV.
- **Per-pixel statistic image:** mean, standard deviation, range, minimum, maximum, median or coefficient of variation across the bands, shown as an image.

Footer (all pages)

- **Applied steps:** the list of steps applied from this tab.
- **Undo Last Step:** reverses the most recent Preprocessing step, including its entry in the shared history.
- **Plot Before / After Spectra:** sample spectra and the current ROI mean before and after the last step.
- **Export Active Cube...:** saves the current cube as ENVI (`.hdr` plus `.img` , float32, band-sequential) or NumPy (`.npz`). The file records the data stage and the steps applied, and the wavelengths in nm if the cube has them.

7. Band Management tab

Band Management changes which bands the cube has. It works on cubes from any file format; the tools that depend on wavelengths offer to assign them first. Every action is added to the processing history.

0. Wavelengths (any file format)

The first column says whether the cube has wavelengths. Click Assign / replace wavelengths to open a dialog with three sources:

- **Uniform range:** a first and last wavelength, spread evenly over all bands.
- **From file:** a `.hdr`, `.txt`, `.csv` or `.wvl` file with one wavelength per band.
- **Paste a list:** one wavelength per band, separated by commas, spaces or new lines.

The count must equal the number of bands. Choose the unit of your values (nm, µm, Å, cm^{-1} or m); they are converted to nm.

1. Remove Spectral Bands

Enter inclusive wavelength intervals in nm, separated by semicolons, for example `410 450; 620 850`. Remove Bands deletes every band inside any interval. Plot Original vs New Wavelengths compares the grids before and after.

2. Resample Spectral Bands

Set a target spacing in nm (a whole number) and an interpolation method: nearest, bilinear or cubic. Every pixel spectrum is interpolated onto a uniform grid over the cube's range. Cubic needs at least four distinct wavelengths. Plot Original vs Resampled Grid shows the result.

3. Select Informative Bands

Set how many endmembers to estimate and how many bands to keep. DS3 finds representative pixel spectra with a fast projection method, scores each band by the contrast between them, and keeps the highest-scoring bands in wavelength order. Plot Selected Wavelengths shows which bands were kept.

After any of these actions the cube has fewer or different bands. Use Revert to original cube to go back.

8. Data Correction tab

The tab has four panels of equal width, in the order they are normally used: radiometric calibration, atmospheric correction, spectral (smile) correction and noise reduction. Each panel has an "About this method" line that opens the description and the paper reference. Every correction that changes the cube is added to the processing history, and the panel labelled "Current data stage" tracks whether the data are DN, radiance or reflectance.

8.1 Radiometric calibration

Converts raw values to physical units. Choose a function, check the parameters, then click Apply Selected Calibration.

Function	Parameters
DN → Radiance	Gain and Bias, one value per band
DN → Reflectance	RGain and ROffset, one value per band
Radiance → Reflectance	ESUN per band, Earth–Sun distance (AU), Sun elevation (degrees)

The fields are filled from the file's metadata when it has them. For any other format, type or paste the values, separated by commas. Bad bands listed in the metadata are excluded when the list matches the cube. Compare Input / Output Band shows one band before and after the conversion.

8.2 Atmospheric correction

Choose the method in the drop-down; only that method's controls are shown.

Empirical line

Fits reflectance = gain × measured + offset in every band, from reference targets of known reflectance.

1. Draw an ROI on each reference target and add it to the ROI list (ROI Analysis tab).
2. Click Add reference..., pick the ROI, then its field spectrum (CSV or TXT with wavelength and reflectance columns).
3. Set the field reflectance units (fraction 0–1 or percent).
4. Click Apply Empirical Line.

With two or more references the fit is a least-squares line per band; with one reference the offset is 0. The field spectra must cover the cube's wavelengths, so the cube needs wavelengths. Plot gain / offset and Plot corrected vs field spectra check the fit.

Dark pixel subtraction

Subtracts a dark value, the haze or path-radiance contribution, from the cube. Choose the source:

- **Scalar value (all bands):** one number subtracted from every pixel of every band. The default is 500.
- **Dark ROI(s) from the ROI Manager:** the mean spectrum per band.
- **Darkest scene pixels:** a low percentile of each band (0 means the band minimum).
- **From file:** a `.npy` array of any supported shape, or a `.csv` or `.txt` scalar or per-band list.

Negative results are kept unless you tick Clip negative results to 0. After applying, a window shows RGB images of the cube before and after, on the same display scale, with the dark value and scene-mean spectra.

Flat field

Divides every pixel by the mean spectrum of a bright, spectrally flat area; the result is relative reflectance.

- **Area:** a rectangle given as row, column, height and width in pixels (1-based; the default is 1, 1, 10, 10), or the ROIs selected in the ROI Manager.
- **Reference spectrum (optional):** choose an ECOSTRESS or CSV spectrum. DS3 then also shows the spectral-angle distance of every pixel to it, before and after correction, side by side. Low values are stronger matches.
- The result window shows contrast-stretched RGB images of the cube before and after, with the mean spectra.

IARR

Internal Average Relative Reflectance divides every pixel by the mean spectrum of the whole scene. It needs no ROI or field data, and the result is relative reflectance.

8.3 Spectral correction (smile)

Reduces the smile effect, the across-track shift of band centres.

- **SpectralSmoothing:** averages each spectrum over a moving window along the spectral axis. Set the window in bands (default 5). Windows above 9 can wash out fine absorption features, and DS3 asks before using one.
- **MNF:** rebuilds the cube from its leading MNF components. Set how many to keep. This is an approximation of the MATLAB method; try several values.
- **Check bands:** two band numbers (1-based, default 40 and 42). After applying, a window shows the absolute difference of those bands before and after correction. With less smile the difference image shows less gradient.

8.4 Noise reduction (NGmeet)

Non-local meets global denoising: a spectral low-rank projection, non-local spatial filtering of the eigen-images, inverse projection and iterative regularisation. The cube is normalised to the range 0–1 first.

Setting	Meaning	Default
Sigma (smoothing)	Strength of the non-local filter	0.01
Iterations	Number of regularisation passes	2
Lambda	Regularisation weight between the denoised and the noisy cube	0.9
Subspace rank	Number of spectral components; Auto lets DS3 choose	Auto
Noise std, Noise seed	For validation mode only	0.05, fixed

Two modes:

- **Validate:** adds seeded Gaussian noise to the normalised cube, denoises it and compares with the clean original. The cube is not changed. The comparison shows RGB images of the original, noisy and denoised data, with PSNR and SSIM.
- **Denoise the active cube:** denoises the real cube and replaces it, mapped back to the original units. It can be reverted.

Sigma works best when it is close to the actual noise level. Show last comparison re-opens the latest result.

9. ROI Analysis tab

Regions of interest (ROIs) are the building blocks for atmospheric references, endmembers and training data. The tab has three groups.

ROI Tools

1. Tick Enable ROI selection.
2. Choose the shape: Rectangle (drag) or Polygon (click points).
3. Draw on the image. The ROI line shows what is selected.
4. Type or pick a class name in Class (for example "Water").

5. Click Add selection to ROI list.

Add current ROI mean as endmember sends the mean spectrum of the current selection to the linear-unmixing endmember list.

Anomaly Detection

Compute anomaly map shows the L2 distance of every pixel spectrum from the global mean spectrum, stretched between the 2nd and 98th percentiles. Bright pixels differ most from the scene average.

ROI Manager

Control	What it does
ROI list	Shows each ROI with its class; the header counts ROIs and classes
Show outlines	Draws the ROI outlines on the image
Delete selected / Clear all	Removes ROIs
Set class…	Changes the class of the selected ROIs
Plot ROI spectra / Plot class means	Plots mean spectra per ROI, or per class
Save ROIs… / Load ROIs…	Stores the ROI list in a file, so it can be reused (also under the File menu)

Flat-field and dark-pixel areas can use the ROIs you select in the list, so select them first. Empirical line asks you to pick a saved ROI for each reference. The ROI-only SNR option uses every ROI in the list, and Classification and the CNN use every ROI, grouped by class.

10. Spectral Analysis tab

Three panels of equal width: Band Math, Clustering and Signal-to-Noise Ratio.

Band Math

Type an expression that uses bands as `B0`, `B1`, and so on, for example `(B80 - B40) / (B80 + B40 + 1e-6)`, then click Apply expression. The result is shown as an image. Band numbers start at 0.

Clustering (k-means)

Set the number of clusters K (2 to 20) and what to cluster on: full spectra, or the first three principal components. Run k-means clustering shows the cluster map. Needs scikit-learn.

Signal-to-Noise Ratio

- **Compute SNR:** spatial SNR is mean over standard deviation per band; spectral SNR is mean over standard deviation across each pixel's spectrum.
- **ROI only:** tick the box to compute the spatial SNR from the ROI pixels, which should be homogeneous areas.
- **Show SNR vs wavelength:** the per-band curve.
- **Show spectral SNR map:** the per-pixel image.
- **Save SNR curve (CSV):** exports the curve.

11. Unmixing tab

Four equal columns that follow the workflow from left to right: estimate how many endmembers there are, extract them, then unmix. Columns 3 and 4 can also be used on their own.

11.1 Estimate Number of Endmembers (HFC / NWHFC)

HFC (Harsanyi–Farrand–Chang) compares the eigenvalues of the scene's correlation and covariance matrices. Each index where the correlation eigenvalue is clearly larger counts as one signal source.

- **False-alarm probability:** default 0.001. Lower values are stricter and give fewer endmembers; values such as 1e-7 are common for large scenes.
- **Noise whitening (NWHFC):** ticked by default. Untick it for plain HFC.
- **Estimate number of endmembers:** runs in the background and shows the result. It also fills the endmember counts of N-FINDR, PPI, FIPPI and NMF.
- **Plot HFC eigenvalue test:** shows the correlation and covariance eigenvalues.

Treat the number as a guide. Compare HFC with NWHFC and edit the counts by hand if needed.

11.2 Extract Endmembers

Pick a method in the drop-down. Each one finds pixels from the image to use as endmember spectra, and plots them against Band Number.

Method	Settings	How it picks pixels
N-FINDR	Number of endmembers; max iterations (default 3)	Pixels spanning the largest simplex volume
Pixel Purity Index (PPI)	Number of endmembers; number of skewers (default 104); MNF or PCA; random seed	Pixels that are extreme most often when projected on random vectors
Fast Iterative PPI (FIPPI)	Number of endmembers; MNF or PCA; max iterations (default 50)	Starts from ATGP skewers and repeats until the skewer set stops changing

After a run, click Use as endmembers for linear unmixing to add the spectra to the list in column 3. The extraction methods work on up to 200,000 randomly sampled pixels, so large cubes stay responsive.

Practical notes:

- MNF preprocessing estimates noise from neighbouring pixels, so it works best on real images, where neighbours are similar. Choose PCA if the pixels have no spatial order.
- PPI uses random skewers: the same seed gives the same result, another seed can differ slightly. It can miss a weak vertex that FIPPI or N-FINDR find, so compare methods.

11.3 Linear Unmixing (ROI / ASD)

Uses the endmember list shown at the top of the column.

1. Build the list: from ROIs (ROI Analysis tab, Add current ROI mean as endmember), from imported ASD spectra, or from column 2.
2. Click Run unmixing (dominant map) to produce a map of the dominant endmember per pixel.

Other controls: Clear endmembers; Import ASD field spectra; Add ASD spectra as endmembers; Plot ASD spectra; Save ASD (CSV).

11.4 Automatic Unmixing (NMF)

Non-negative matrix factorisation finds endmembers and abundances without a list.

- **Number:** how many endmembers (2 to 100).

- **Seed:** NMF is not unique; another seed can give another equally valid result (default 42).
- **Bands:** the band range to use.
- **Fit px:** endmembers are fitted on a random subset of pixels (default 100,000; 0 means all), then abundances are computed for every pixel.
- **Run NMF unmixing**, then **Show** one abundance map, **All maps** in a window, or **Endmember spectra**.

NMF needs scikit-learn. Negative values are clipped to 0 and the info panel says how many.

12. PCA, Classification and 2D CNN tabs

12.1 PCA tab

PCA / Dimensionality Reduction

Set how many principal components to compute (default 6) and click Compute PCA. Then:

- **Show PC1, PC2, PC3:** each component as an image.
- **Show PCA RGB:** PC1, PC2 and PC3 as the red, green and blue channels.
- **Show Scree Plot:** how much variance each component explains.

Computed components can be reused as features in the Classification tab and for k-means clustering on the Spectral Analysis tab.

Spectral Angle Mapper (SAM)

Classifies each pixel by the smallest angle between its spectrum and a set of reference spectra.

- **Reference spectra:** the mean per ROI class, or the endmember list from the Unmixing tab.
- **Max angle (degrees):** pixels whose best angle is larger stay unclassified; 0 means no limit.
- **Run SAM classification** shows the class map; **Show best-match angle map** shows how close each pixel is to its best match.

12.2 Classification tab

Supervised classification trained on your labelled ROIs. Create ROIs with class names first (section 9).

Setting	Options
Classifier	Random forest; SVM (RBF kernel); SVM (PCA + poly kernel)
Features	Full spectra, or PCA components (compute them in the PCA tab first)
Validation split	Hold out whole ROIs (honest test on regions the model never saw), or random pixels
Test fraction	0 to 50 %, default 30
Max pixels per class	Caps training size, default 5,000

Train and classify image trains the model, classifies the whole image and shows the class map. Show accuracy report opens the per-class results. Run k-fold cross-validation (accuracy only) repeats the training in 2 to 10 folds (default 5) to estimate accuracy without classifying the image.

Holding out whole ROIs usually gives lower but more realistic accuracy than random pixels, because neighbouring pixels are very similar. Needs scikit-learn.

12.3 2D CNN tab

A convolutional network trained on small image patches taken from the PCA of the cube and your labelled ROIs. Needs TensorFlow.

Setting	Meaning	Default
PCA components (whitened)	Channels of each patch	30
Patch size (odd)	Side of each patch in pixels, 9 to 31	15
Epochs	Training passes	20
Batch size	Patches per step	64

Setting	Meaning	Default
Network	Reference (64-128-256-512) or Compact (32-64-128, faster)	Reference
Validation split	Hold out whole ROIs or random pixels	Hold out ROIs
Test fraction	Share held back for testing, 0 to 50 %	30
Max patches per class	Caps training size	3,000
Classify	Whole image (slow on large images) or labelled ROI pixels only	Whole image

Click Train 2D CNN and classify, and watch progress in the info panel. Stop task cancels training. Show report / training log reopens the results. For a first try on a large image, choose the Compact network and classify labelled ROI pixels only.

13. Processing history, revert, saving and exporting

Processing history

Every operation that changes the cube, on any tab, is added to one shared history: Preprocessing steps, Band Management actions, radiometric and atmospheric corrections, smile reduction and NGmeet denoising. Each entry records the step name, its parameters, the band count before and after, the data stage, and the time. The line in the history bar shows the count and the latest step. Show history (or View → Processing history) lists all of it.

Save history… writes the list to a JSON or text file, so you can document how a result was produced.

Revert to original cube

The button in the history bar (and View → Revert to original cube) discards every processing step on every tab. It restores the cube, its wavelengths and metadata exactly as loaded, empties the history, and resets the data stage to "DN / unknown". It asks for confirmation first. Analysis results such as PCA and classification are cleared; ROIs are kept. Because the original is kept in memory, reverting is instant.

For a single step, use Undo Last Step on the Preprocessing tab; it also removes that step from the history.

Data stage

DS3 tracks what the values mean: DN / unknown, radiance, TOA reflectance, relative reflectance (flat field or IARR) or surface reflectance (empirical line). The Data Correction tab shows it, and it is saved with exports. DS3 warns before applying a relative-reflectance method to data that are already reflectance.

What you can save

Item	Where	Format
Processed cube	Preprocessing → Export Active Cube	ENVI (.hdr + .img, float32, band-sequential) or NumPy (.npz)
Processing history	History bar → Save history	JSON or text
ROIs	File → Save ROIs / Load ROIs, or ROI Manager	ROI file
Spectral index image	Preprocessing → Indices & Features → Save Image	Image file
Band statistics	Preprocessing → Statistics → Save Statistics	CSV
SNR curve	Spectral Analysis → Save SNR curve	CSV
Imported ASD spectra	Unmixing → Save ASD (CSV)	CSV
Any plot window	the toolbar of the plot window	Image

The exported ENVI header carries the wavelengths in nm, a description with the data stage and the list of steps applied, and NaN as the ignore value.

14. Typical workflows

A. Raw DN data to surface reflectance

1. Open the cube and check the wavelength range in the info panel.
2. Data Correction → Radiometric calibration: convert DN to radiance, or DN to reflectance, using the gain and bias in the file or values you type.

3. Atmospheric correction → Dark pixel subtraction to remove haze, if needed.
4. Draw ROIs on reference targets, then Empirical line with their field spectra. The data stage becomes surface reflectance.
5. Check the corrected spectra with Plot corrected vs field spectra, then export the cube from the Preprocessing tab.

B. Clean a noisy cube

1. Spectral Analysis → Compute SNR and Show SNR vs wavelength to see which bands are noisy.
2. Band Management → Remove Spectral Bands for the water-absorption regions.
3. Data Correction → Smile reduction, then check the two-band difference image.
4. Noise reduction (NGmeet) in Validate mode first to choose Sigma, then in Denoise mode on the real cube.
5. Compute SNR again to compare.

C. Match a scene to a library spectrum

1. Draw a flat-field ROI, or enter the rectangle, and choose a reference spectrum file such as an ECOSTRESS vegetation spectrum.
2. Apply Flat field. The result window shows the original and corrected spectral-angle score maps side by side.
3. Low scores mark pixels close to the reference. Use Revert to original cube to try a different area.

D. Find the materials in a scene without a library

1. Unmixing → column 1: Estimate number of endmembers (compare HFC and NWHFC).
2. Column 2: run N-FINDR, PPI or FIPPI with that number. Compare the spectra.
3. Use as endmembers for linear unmixing, then Run unmixing in column 3 for the dominant-endmember map.
4. Or run NMF in column 4 for abundance maps for every endmember.

E. Supervised classification

1. ROI Analysis: draw ROIs for each material and give each a class name.
2. Optional: PCA tab → Compute PCA, then choose PCA components as features.
3. Classification → choose a classifier, keep Hold out whole ROIs, then Train and classify image.
4. Show accuracy report; run k-fold cross-validation for a second opinion.

F. A file with no wavelengths

1. Open the file and answer the axis prompt if one appears.
2. Band Management → Assign / replace wavelengths from a list, a file or a uniform range.
3. Wavelength-based tools (band removal, resampling, indices, empirical line, spectral matching) now work.

15. Troubleshooting, limitations and planned features

Common problems

Symptom	Likely cause	What to do
A tool says it needs wavelengths	The file has no wavelength information	Band Management → Assign / replace wavelengths
Wrong spectral axis, or a strange image	A 3-D array with an ambiguous band axis	Reopen the file and pick the correct layout in the axis dialog
"Not enough memory" when opening	The cube is larger than free RAM	Open a spatial subset of the file, or use a machine with more memory
A format will not open	Its optional package is missing	Check Help → Supported file formats and install the package
k-means, NMF or a classifier shows a message	scikit-learn is missing	Install scikit-learn
The 2D CNN tab does nothing	TensorFlow is missing	Install TensorFlow
Denoising or training is slow	Large cube or many epochs	Use the Compact network, classify ROI pixels only, or press Stop task
Wavelengths look like 0.4–2.5	Values are in µm	Preprocessing → Units → Convert Wavelengths to nm
A result looks wrong after several steps	An earlier step was not suitable	Open Show history, then Revert to original cube and repeat with the right order

Symptom	Likely cause	What to do
Results from a long task vanish	A different image was loaded while it ran	Wait for tasks to finish before opening another file

Things to know

- The whole cube is held in memory; large scenes are stored as float32.
- The MNF smile reduction and the Fast Iterative PPI final selection are DS3's own interpretations of the published methods. They may differ slightly from other software, so cross-check important results.
- HFC and NWHFC give estimates. They can differ from each other, and a different false-alarm probability changes the count.
- PPI is random: use the same seed to repeat a run exactly.
- Dark pixel subtraction keeps negative values unless you tick the clip option.

Not yet available

- A spectral library database with material categories and multi-scale entries.
- Dark and white reference calibration for lab and field cameras.
- Convolution of spectra to a target sensor's spectral response.
- Georeferencing, and aggregation of several scenes into one analysis.
- Preprocessing chains saved with each spectrum (the history is saved per cube today).

16. Appendix

Glossary

Term	Meaning
Cube	The hyperspectral image: rows × columns × bands
DN	Digital number, the raw sensor value
TOA	Top of atmosphere
ROI	Region of interest drawn on the image
Endmember	A pure material spectrum that other pixels are mixtures of

Term	Meaning
Abundance	The share of an endmember in a pixel
HFC / NWHFC	Harsanyi–Farrand–Chang test for the number of signal sources, without or with noise whitening
MNF	Maximum noise fraction transform: PCA after whitening the noise
PPI	Pixel purity index: counts how often a pixel is extreme on random projections
FIPPI	Fast iterative PPI, started from ATGP skewers
ATGP	Automatic target generation process: picks pixels with the largest residual in turn
SAM	Spectral angle mapper: classifies by the angle between spectra
SNR	Signal-to-noise ratio
PSNR / SSIM	Image-quality scores used to compare a denoised cube with the clean one
Smile	Across-track shift of band centres in pushbroom sensors
Relative reflectance	Reflectance up to an unknown scene-wide factor (flat field, IARR)

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Application: DS3 version 1.0, created by the Kapih Team.